

ActInf Textbook Group ~ Cohort 2 ~ Meeting 11 (Chapter 5, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 11 (Chapter 5, part 2) | 1:01:16

hello everyone thanks for joining or
listening it's November 11th
11 11 22.
and we're in cohort two of the active
textbook group
having a discussion of chapter five
exploring some topics
there are some questions
that have been written in comments as
well
and then
taking it wherever we want as we review
the first half of the book
that'll be today
and then
next week we'll be taking an even
further step back
and more of a focus on like feedback
completing the feedback form
how can we improve and structure future
textbook group cohorts project ideas and
so on
but for today we're in chapter five
so
um
does anyone want to
give any thoughts or comments or just
initial questions to raise about chapter
five
all right then let's go into the written
questions
okay
how does the simple hierarchy with
one-to-one mapping between levels as
shown in the left hand side of figures 4

4

6.

and 5.2

44

four six

five two

relate to actual brain hierarchies such
as the classic fellerman and Van Essen
diagram

from retinal ganglion cells up to
hippocampus with many to many mapping
between Regional units
ah

I guess this is the famous
diagram

well what do people think how do these
seemingly
skyscraper like

hierarchical models

this one is laid out side to side but we
can see ascending and descending so this
could have been graphically laid out
like this and then another one and then
a third one on top
of course top down the bottom-up are
just spatial metaphors

so

um

how do we reconcile

computational hierarchical models
with

the empirical structural connectivity of
the brain

which

has

a wiring structure

looking like this

Ali

between these two kinds of icons or kinds of maps because on one hand we have a kind of relational map and on the other hand there is a kind of architectural slash functional mapping of the brain so for example if we look at a circuit map of the of an electric circuit there's we have a relational map there right but when we want to implement that circuit I mean in terms of wiring and so on we need to have another kind of map which is architectural one so of course in some cases there's a possibility of translating one map perfectly onto the other but there isn't necessarily an isomorphism between those two kinds of map and I think in this case we don't necessarily see that isomorphism here as well excellent great answer totally agreed these kinds of diagrams are often described as hierarchical or heterotical or people use difference um words to describe it but it's describing the anatomical connectivity of different brain regions whereas the diagrams that are shown in these figures that were mentioned in the question are not anatomical Regions they're statistical variables connected by edges this is the difference between structural and effective and functional connectivity structural connectivity is

what pieces are actually put together

and through direct contact

intermediating

whereas

functional connectivity has to do with

the statistical relationships in the

system

and those are not necessarily for a

given set of measurements or spatial or

temporal scale or experimental stimuli

those are not exactly the same thing

there are cases where there are but that

doesn't mean that in general they're the

same thing so how do they relate to

quote actual brain hierarchies

they relate to each other in the way

that Ali just described

similarly between

um uh architectural diagram and a

functional diagram

does anyone have any other

thoughts on this

yeah Giuseppe

yeah I think that is a is a very good

explanation on the other hand I think

that at that stage of the of the

description the idea was to see how

active inference could also provide a

process Theory so basically to to see

how these information mechanism may be

implemented in a neural circuits so

yes the two are not completely the two

descriptions are not isomorphic but I

think there should be some degree of

of uh agreement and

in this case I think that

what may be masking the the the

similarities or the the potential

overlap is that you know there there's a
a big difference in the degree of
complexity of the so one is a very
simple the the active inference diagram
is a very simple sketch of a very simple
micro circuit while this one is a big
architecture with a lot of complexity
and it's difficult to
to go in and say you know
to map exactly those kind of micro
circuits within this large circuit
scheme
it's probably the there's going to be
various
micro circuits in there with the various
degrees of a hierarchical structure
within that
that large scheme I guess
yeah
um all great points
the figures that were pointed to
reflect like just
simple
illustrative examples
whereas the whole brain
or many of the regions of the brain if
one were to even just describe this with
an active inference model it wouldn't
look as simple and narrow as these
previous figures so it's a different
type of diagram it's a different scale
yet
through Visual and prose-based
argumentation
as well as the the hopes and dreams of a
field
is that there is a relationship between
the messages that are passed between

different variables statistically which
as we discussed last time it can you
could arrange these in a hexagon
it wouldn't change the topology and thus
the function of the base graph
however it is being laid out in a
specific way
to demonstrate resonance with
given cellular populations in a cortical
column
so it's kind of
um
situational to what extent the
structural neuroanatomy
aligns with the Bayes graph as specified
but
we can definitely say that they're not
the same thing
that they don't have to be the same
thing
but there's also situations where by
graphical design
using that in multiple ways meaning you
could constrain your construction of the
base graph
such that it has some structural
patterns with the anatomy and then also
just in terms of like visual graphical
design
like you can just lay it out visually
because this is not these are it's not
exactly like there's five and then three
like there's other things that are here
that aren't shown here
so like this is a a course mapping
any other thoughts on this question
asked
uh just really briefly I feel like the

30 million foot view of the like
structural constraints sort of is that
in the general case of a base graph your
well inactive inference we're talking
about like a Markov chain or some
blanket that is kind of ultimately the
state space kind of which is included in
it
um
if you're trying to map that to a brain
um
or any kind of
system with discrete units that are
close sometimes and not so close then
the arrangement of those will constrain
necessarily the base graph
to some degree
if you if they're right next to each
other
then there's going to be a greater
chance that they are
Independence
some sort of dependency relationship
even if the from like disparate
you know far away neurons are actually
more strongly correlated they will have
to travel through those in some sense
the the network there right
um so
I'm not sure if that's clear but the
what I'm saying is that the physical
proximity and adjacencies in a physical
system kind of
partially correlate to something like a
markup blanket like the constraints the
possible operation in the system
yes it can
and also this was a major Topic in the

recent

51 live streams

like we talked about how different by

like the pomdp structure partially

observable more composition process

structure or just generally the active

Bayesian graph representation of a

generative model

that's not

the anatomy that would carry out that

model

you would need to make a neural network

with a totally different connectivity to

implement that model

and so we discussed how do you get

from

a stated base graph

to a neural network architecture that

carries out that computation

which helps integrate it more closely

with experimental results and also

software packages and different kinds of

computational heuristics

and then how do you get from a neural

network

to a pomdp

and so this Paper 51

establishes this relationship

formally as well as empirically

so it's

extremely important reading slash

listening in the space of beyond the um

more foundational points that have been

made in the last several minutes which

is just simply

these kinds of graphs

are not

these kinds of graphs

that's just two different categories of visual representations that both use nodes and edges but they're describing different things with nodes and edges and then to go deeper and to understand exactly how biological and artificial neural architectures

implements the kinds of generative models specified by a Bayesian graph then the paper discussed in 51 is the most recent and advanced discussion on it in the field

okay

how are categories actually encoded in the brain

so can you can you add that reference

there oh sure

thank you

yeah

and they've done work in in silico

in vitro neurons and in Vivo animals it

seems like

so

funny there's fef free energy functional

but it's definitely not that

and there's been other

discussions of just structural versus

effective and functional connectivity in

the brain

all right so how are categories actually

encoded in the brain and Neil added a

comment

I understand that prediction errors are

thought to be encoded as pulse density

I.E frequency modulated Spike trains as

measured on single neurons

this might lead us to think that

categories are encoded in single neurons

like the notorious grandmother and Jennifer Aniston neurons which are neurons that have been reported in different studies to have an increased and sometimes like specific firing rate increase for a specific stimuli so they'll have they'll be measuring thousands of neurons and some neurons fire when some image is brighter or darker and some neurons don't have any clear relationship with visual input but then they'll find some neuron that fires for Jennifer Aniston but not for other closely related stimuli and so hence the fame of the Jennifer Aniston neuron in this schema are we really saying one as far as active is concerned it doesn't matter how categories are stored within a cortical column components we just need to model that they are with associate precisions two we suppose categories are stored statistically somehow collectively within a local population of neurons Would we not expect prediction errors to be represented in Spike trains on many neurons coming into or going out of a cortical column all right there's many ways to explore this question who wants to give a first thought wait on you Ali yeah go for it Ellie oh well actually it's a very big question in the philosophy of mind and the philosophy for cognitive science I mean the question of how uh brain represents the uh the signs and the external objects but um

one theory is
um basically based on versus the
taxonomy of signs uh I mean as we
um as many were aware uh pursue
classifies or the signs into three types
of symbols icons and indices uh but
when we want when we
apply this taxonomy
in the representation mechanism of brain
basically what we need at the very first
as as a very first step is to understand
how the brain represents indices and
icons
as a kind of
representation processes that would need
minimal amount of Consciousness right so
uh for indices and icons uh I mean the
difference between indices and I icons
are that icons are a kind of uh
structural structure preserving maps and
indices uh are uh we in indices we have
a kind of causal relationship between
the signs and their reference so before
uh
um before uh we
um
representing uh the tokens of an object
as an index of an external object we
need to have a kind of map or an icon in
order to map that in that token of index
to our icon right so one theory for
doing that is so-called homuncular
theory proposed by Daniel Dennett in
which he proposes a hierarchy of
subjects with decreasing levels of
consciousness
and on the bottom most level We have
basically non-conscious non-conscious

subject to somehow grasps the icon in
a kind of just a
representational map without uh
interpreting it as a specific
um as a specific Index right but then
again uh it passes that representation
into its higher levels of hormone
homunculus uh with uh increasing their
increasing Consciousness until it
reaches the subject quote-unquote
subject which is basically our whole
mind so yeah it's a very big subject and
of course

it's impossible to summarize it in just
a couple of minutes but I would
recommend a very interesting book on
that subject if anyone wants to further
explore that the the first chapter of
materialist phenomenology by Manuel
enter

explains it with a crystal clear prose
what that kind of representation or how
that mechanism would work
nice

great answers so a few more
angles on this question
um

first the question of representation in
active and in Neuroscience is a large
area
so

for something to be represented
in the brain

it doesn't need to be what it is in the
outside world clearly but that leads to
like a lot of of course

um discussion about how representations
actually do occur in the brain

so the question

um was prediction errors are encoded as
pulse density as measured on single
neurons well the simple answer is it
just simply does not have to be that way
prediction errors in a different
cognitive system or region could be
encoded as the autocorrelation function
of the spike train

or there are neurons that are
non-spiking

or it could be a brain region there's
just no reason to uniquely associate the
conveyance of prediction errors
to the extent that statistical
parameters are even conveyed in
neuroanatomy at all which one doesn't
even have to agree with but to the
extent we're even going to talk about
neural phenomena as being related to to
statistical phenomena

it doesn't have to be the spiking
density

nor does it have to be on single neurons
now that being said
in the neural network
discussion in 51.

there was a relationship that they are
showing between the fast perceptual like
Dynamics and Spike firing rates and the
slower Dynamics and the neural
plasticity of the synapses so as kind of
before like it could be but just because
it could be in in some senses maybe
doesn't mean that those are an identical
mapping

um

then

categories are encoded in single neurons
um well if by that is meant encoded by
firing sensitivities in networks of
neurons

then there is already more being taken
into account it's kind of like figure
and ground it's not like that neuron
taken out of that Network would still
signal Jennifer Aniston the semantics of
a given category

are associated with the inputs and
outputs even in cases where there is a
statistically tight relationship between
the firing activity of some neuron and
some other category being identified
um then

are we really saying
it doesn't matter how categories are
stored we just need a model that they
are so that is an instrumentalist
perspective

which is just saying
whatever the neurological substrate is
of

how categories are enacted and
represented in the brain

we're going to use a specific type of
model

that is super tractable supercomposable
translates across systems and so on
hashtag active

and we're not saying that's the
architecture of the system

so that's an easy instrumentalist
position that can always be taken

and we suppose that categories are
stored statistically somehow
collectively within a local population

of neurons that's an empirical hypothesis one could test whether a population rate code or a area of activity or other aspects of neural function that aren't even as commonly discussed depending on the specific question come into play

Would we not expect prediction errors to be represented on Spike trains on many neurons coming into and going out of if you expect it if that's your prior then you expect it if it's not your prior you don't expect it there might be anatomical or statistical reasons

to scaffold the belief that there would be a many input many output system or not but it can't be answered in the general those are just the examples of Empirical neuroanatomical research to actually pursue

um Brock how do you see it related to viveki

um I mean relevance realization is uh

uh I mean very roughly like

opponent processing it's I mean it's

I don't I don't really

understand the fundamentals of precisely

how that would work in a

neurological physiological sense

um

but it's kind of again like I guess you

could uh like

yeah the promotions there

um but

um yeah that just that you're
contrasting like different
um fittedness I guess of um
in in the live stream he gave this I
guess example of um like a dog versus a
four-legged animal versus a mammal or
whatever why did they say the dog ran by
sort of thing

um is that that has a good fit
um

and that
realized relevance is somehow related to
opponent processing
um so it's it's that's I guess a way in
which it's related to categories or very
crudely like

General differentiation or something
like this

um
tangentially related I don't know yeah I
don't know it's like a conceptual
approach too yeah there's always
different coarse grainings of categories
that can be appealed to
in speech or in the brain
and so outside of contrived
you know sorting tasks
in the more open-endedness of the niche
that the systems have to engage with
the categorization

[Music]

is has different characteristics
than like a binary sorting task
so that's that's what viviki ties to
relevance

okay

does anyone want to add another question
or thought on five

what other neural systems or behaviors
are people interested in modeling
foreign
but also in that in that uh diagram
behind the
window there there's goals
Golds yeah where did those come from
seems like um
I guess you could do planning without
goals but it seems like goals a goal
there would be a
kind of a reason to plan to get a
biological goal of like
homeostasis right Isasis is sort of this
sort of
there was a live stream you guys did
recently on that right
and then
some more subjective cultural thing yeah
I don't know
yeah well here
um with the more narrow
goal interpretation
the difference is between the direct and
the indirect policy selection Pathways
in the stratum which is mediated by
gamma
the dopamine signaling
and so here on this left side
the habits dominate policy selection
so e represents the affordances but it's
not just like affordances are not just
like yes no in the E vector
they represent
the prior probabilities of given actions
so where an affordance is impossible it
is given a prior probability of zero and
therefore it's never selected

where the prior probability on some
action affordance being engaged in is
high

it's more likely to be undertaken so e
is not just like a list

of what you can choose from it's not
like a menu It's like a menu of actions
that's weighted by habit

in this Construction

and so

in the absence

of any kind of pragmatic or epistemic
value which is to say free energy
minimization

Behavior recapitulates The Habit

so there's a mode of policy selection π_i

where the entity is merely

recapitulating

probabilistically

their history and habits

that's shown on the left side of the
stratum

then the right side of the stratum

shows the cortical engagement

of

seemingly more goal directed whether
pragmatic goal directed and or epistemic
goal directed

or as we think about it in active

inference then the pragmatic and
epistemic integration

into a single imperative G

like

um let's just say I usually walk up and
down but I rarely go left and right and
I'm in a maze

this left mode

of policy selection is just going to be

recapitulating the decisions previously made

the right mode is going to ideally strike a balance between pragmatic and epistemic value

but any discussion of pragmatic and epistemic value so pragmatic value bringing alignments reducing the Divergence between your preferences and observations

getting what you want getting what you expect

an epistemic value reducing your uncertainty gaining information

any discussion of the relative merits of different policies

in terms of their expected free energy which again is to say they're relative or

relative pragmatic and epistemic value

any evaluation of policy is a different cognitive and computational process

than merely the recapitulation of habits which doesn't require any consideration of policies at all

so this is kind of like walking forward walking into the future looking backwards

I mean in some deeper sense aren't we always

but this e-mediated mode of decision making

does not require any evaluation of counterfactuals or comparison of alternative outcomes

now where do habits and preferences come from

because if under G the pragmatic value
is going to be associated with
prior probabilities on preferences
and um epistemic information as well is
going to be mediated by priors which
here are being shown as kind of
cortically embodied
where do those parameters come from
Evolution ecology development and
behavior over different time scales
so the coarse grained priors are
inherited
through the process of evolution
and then finer and finer scale
adjustments can occur
via ecological and developmental
processes
culminating in learning
and at the very fast scale perception
that's the deflationary approach which
would be Evolution has put all those
tested processes Evolution ecology
development Behavior learning and then
perception
put that entity's generative model in
the context of a generative process
into
an Adaptive position
if it's not adaptive that far from
equilibrium system is going to quickly
dissipate you just saw something in its
last day
or it's not its last day and then the
inference is going to continue Giuseppe
yes

[Music]

something that is not too clear to me
habits are supposedly supposedly they do

refer

uh to things that are learned during the
lifetime of the agent

um

as opposed to being a

deposited through Evolution for instance

because one could say also that the the

the policies with high pragmatic values

that that are the one that preserve the

phenotype of the agent are kind of

habits but they pertain to the longer

time scale of uh phylogenesis so to say

so

so is there

the distinction here

like the specific of habits the

characteristic of web is is that they

have been learned they they are they

come from a statistical regularities

that have been uh accrued and

represented through the lifetime of the

agent to the experience of the of the

agent

I believe that it doesn't have to be

only learned during the lifetime like

let's just say that there's a species of

bird that flies at night

it habitually flies at night

and then you can zoom in more and say

now within that bird species they're the

early risers and the late risers

and this one is habitually Rising early

and that one's habitually Rising late

now that may be a genetic polymorphism

that doesn't even have to be learned at

all

so I think of course in humans we take a

very developmental learning based

approach towards the habits and you know
Aristotle you know Excellence is what we
repeatedly do and all of that but when
we just think about habit as what is
habitually done then it's a little bit
clearer that it's not necessarily
learned
but then once you couldn't you
subsume the
like the the tendency to fly more at
night as a as a preference that is
within
the
[Music]
within the
yeah as a preference like within the
same Matrix
yes and in some
um convergent evolution scenario
we would see the Habit distribution
recapitulating the preference
distribution increasingly mindlessly
offloading the cognitive work of
evaluating policies
and reducing Divergence through free
energy minimization
we might expect some of that
to become habitualized
and merely reflected through the
affordances that can be taken or the
probability of different affordances in
terms of their prior preferences or
prior probabilities on affordances
um rather than needing to be mediated
through preference that's one answer and
another kind of corollary answer is on
one hand we can talk about preferences
over any distribution

but in the bird example

um

the preference for flying at night might
not be a category that's accessible to
it

but

preferences

that have to do with um

the amount of photons being detected
might interact synergistically with
habits around like a circadian rhythm or
habits around flapping your wings
like once the wing is flapped up it
always flaps down

that doesn't even need to be
computationally modeled as a policy
decision

because it could be relying on these
regions of the brain or these kinds of
pathways

that don't explicitly consider
alternatives in the same way

and I think again as we discussed at the
beginning this isn't the literal
interpretation of the neuroanatomy but
it's kind of showing two extreme
um ends of a continuum

with

um the least to the most explicit
consideration

of future policies

with all the pros and cons of turning
the volume up turning the volume down
and the volume is computationally
analogous to the role of dopamine
signaling in the striatum

and in insects Ollie

uh yeah and uh another slightly

different take on this subject is laid out by J Scott Kelso in his book Dynamic patterns the self-organization of brain and behavior uh in which uh he says that because uh the state of uh all the possibilities are too complex to be calculated at each um at each time and for each decision

he uh he lays out something like this that if we look at it and look at the state of the possibilities

um in terms of its uh distribution of singularities and the basins and The Well of those singularities or or the attractors of uh the Basin of attractors the policy making can be achieved by uh by the effect of the param by the effect of adjusting the parameter and the proximity of those Wells or basins so for example

manipulating that parameter parameter to make some wells or basins shallow with the result that the associated preferred state becomes less stable so in this condition random fluctuations actually can spontaneously somehow dislodge the system from a data space that estate and push it into another state so in this case

The Habit can also be explained by the varying degrees of these kind of parameter adjustments throughout all the state of possibility States

awesome yeah

yeah this is this is

very interesting one one subject in which I'm

[Music]

I'm interested in applying uh the active inference framework is uh on a meditative practice and actually it has been one proposal is that it may work exactly like that so that uh the the tractors that are very deep and um and are firmly Consolidated through habits through meditative practice becomes shallower also that the range of possibilities of behavior and perception becomes larger because you you you are not constrained by this deep Wells of uh of uh of energy that have been established through your habits but the interesting thing here the Intriguing part is that through a very uh repetitive practice kind of repetitive practice you are you some how make your uh energy landscape more shallow at least at least at least during the practice so it's a it's an interesting this paradoxical aspect how you you could somehow train with the very habitual practice to weaken your habits oh just extremely not of a tangent here but it's totally interesting and one could imagine um real practices or simulated computational agents that engage in repetitive actions and that just sharpens it doesn't open their mind a lot of meditation is not about expanding

one's mind or changing one's Behavior
it's actually about converging to
a certain regime of attention
and sharpening distributions that exist
during the practice and after the
practice

there are other cases where we can
modulate our receptivity
through repetitive actions
and being exposed to just merely novelty
all the time doesn't necessarily
breed novelty in US it might just
confuse us

mm-hmm

yeah

yeah Brock some habits are reciprocally
narrowing some are reciprocally opening

yeah

yeah they can look very similar
and without

um the ability to see the cognitive
states of another

if somebody's just sitting quietly
maybe they're narrowing maybe they're
opening maybe they're
sleeping

well to your point you mean about you
know like some basic version of
meditation of just like focus on your
breathing

was like sharpening and narrowing it but
in doing that

over time you kind of grow this capacity
to modulate your attention more
on other things more intentionally right
um so

right probably not

perhaps this idea that by doing that by

deploying attention more on the sensory
aspect you kind of avoid a little bit or
bias the usual drift of precision
upwards toward more like Superior layers
of the hierarchy associated with mind
wandering Etc

so this

wrong one would you like to add anything

okay

awesome I'm going to copy your comment

in so we can read this

I would put forward here the practice of

Alexander technique where we optimize

Precision by repetitive action I.E

getting in and out of a chair whilst

inhibiting and directing

hmm

yeah it'd be interesting to

[Music]

um

with this three

region neuroanatomical model

yeah

Oracle

the most cognitive most semantically

accessible

the midbrain striatum

most dopamergically associated with this

um Continuum between habitual and goal

directed policy selection and then

action mediated in the peripheral

nervous system through

um Divergence minimization

and then

sketch out

I'm sure

and then sketch out yeah like um in the

Parkinson's case where there's a loss of

dopaminergic signaling
um which can in the extreme lead to like
failures of initiating and planning
action
and then
um there was a
um
thank you
Robin Williams movie
Awakenings
this movie is about a uh early on
Oliver Sacks wrote that book but then
there's a Robin Williams movie
and um there was a floor of patients in
a Medical Institution
who were all suffering from basically
these different symptoms that today
could be recognized as like hypo
dopaminergic
um
signaling and then
um
the doctor goes to a conference learns
about l-dopa which is like a pre a
metabolic precursor of dopamine so
dopamine does not cross the blood-brain
barrier in mammals however l-dopa does
and um
then that like awakens people but it
shows
um
like some of the motor and cognitive
consequences of hypo and Hyper
dopaminergic signaling
um one question for me is where does
reasoning fit into thicker 5.5
the gears of thought
where's the transmission in the

carburetor

like those are great

questions so there's probably many
different

um

angles to take on it

one

angle is like reasoning is a a synonym
of inference

and so by pointing to reasoning

we are talking about

families of inferential structures

and so depending on what that reasoning
is about

it could manifest in different ways

and so like yeah Ali

actually helmholz proposes that

reasoning can as you said it can be it

can be done also unconsciously as

inference so uh actually one of the
starting points of this whole area

Bayesian brain hypothesis and probably

its origin is uh helmholtz essay I

forgot its exact name from 1866 or

something in which he proposes a theory

of unconscious inference or unconscious

uh conclusion uh in which uh sometimes

reasoning actually most of the time the

majority of our reasoning processes are

not done in our conscious levels so uh

yeah I also agree with you that

reasoning can be taken as a synonymous

for these kind of inferences as well

yeah but like on the experiential

side it is so

fascinating and mysterious

stimuli are presented

certainly there are

um unconscious inferences that
facilitate even the initial
understanding of a stimuli
like a visual stimuli like seeing a
letter as a letter
seeing a handle for grabbing is kind of
the ecological inactivist angle
but seeing a symbol or sign and that
kind of invokes what Ali described
earlier with the theory of semiosis
but then
we wait
and we feel like we reason
and we
make decisions
and so how to

[Music]

um
how to square the graphical models
of inference which is to say reasoning
with that yeah Brock
I was just gonna ask I guess where
mental action fits into this on policy
selection or habits or goals like
yeah yeah
like um I yeah I I think the vast
majority of this is yeah that you get
like this base
unconscious inference but at some point
it feels like there's something
conscious there about selecting between
two things sometimes
it really
intentional and
thinking real hard
um so I wonder what that is I guess
yeah
Giuseppe

something about going back to habits
something that I just thought about
um so usually habitual actions from the
point of view from the psychological and
phenomenological point of view are
they seem to require
much less mental energy or psychological
energy so we can think about that in
terms of uh you know the theory of
system one and system two like by then
economy and

I wonder if they're seeing they seem to
have some preferential uh low levels of
of free energy also of expected free
energy uh the police the habitual
policies compared to other policies that
somehow

mirrors there

they're relatively low investment of
energy psychologically

yeah well that's been

a hotly debated topic

and one of the most mystifying
observations

is that the brain does not change in
metabolic activity greatly
depending on the task and so there's all
these amazing analogies between Freud
and free energy and all these things
that Psalms and fristians and others
have worked on and of course the
tantalizing connections between free
energy minimization and energetic
um balance

and the um fmri signal that's being
studied is the brain oxygen levels
changing bold signal and also people can
study the glucose metabolism

but these levels of change
are very small
in brain metabolism and glucose usage
and they're not simply associated with
like quote thinking hard and even
necessarily cognitive demands
and so it's it's like it's like a I
don't know if it's a hammer without a
nail or a nail without a hammer because
empirically it doesn't change in energy
use that much
although although there was a recent
paper
and I think it's in
it's on a big Journal I can't remember
if it's nature or
or actually
it was
um
there was you know they proposed some
[Music]
some kind of uh
approach similar to energy minimization
and and that they were showing that
actually the
the energy
spent in in synaptic communication was
was uh minimized and if I I may look for
it and I may post it here
um
but I wonder more specifically
in this in this idea of that habitual
patterns of behavior or let's say
sequences of action gets shifted with
practice to being process from the
cortex to the basal ganglia
somehow there is uh there is actually
[Music]

well not not only not only like a
metabolic energy like in terms of
glucose or Advantage but also in terms
of uh really free energy so
I don't know if that has been looked
at
uh specifically like for the basal
ganglia or I don't know there must be
something because
it does seem that
uh
mental resources at least experientially
are become much
[Music]
the requirement resources become much
less once
once you a task is automatized let's say
yes just shortly to close this and and
to address this uh
final question when does passive become
active inference and end planning well
we could look at from a base graph
perspective is action involved
if the task is merely perceptual we can
say that it's passive inference
if ocular motor cicading is required so
even if the person's in a chair but
they're moving their eyeballs that could
still be modeled as active inference and
you'd want then a unified imperative
for inference and action
and then when is it planning well as we
had some controversial discussions on in
cohort one about the second half of the
book is it really planning do we really
see planning in this textbook
um as if planning would be behavior that
looks like it's enacting a plan like a

chess player who's winning a game but
the strict and narrow definition of
planning might be explicit consideration
of sequential actions and their
consequences

and preparation for those in evaluation
of them

and free energy reduction or
minimization could be used in purely
passive inference which is where it's
widely used in general

as an Evidence lower bounds and in
machine learning without action in the
loop

so the innovation of active inference is
to bring action into the fold of
inference

and make a unified free energy
functional imperative

for perceptual learning and action
inference

all right

hope it's okay we went a few minutes
over but thanks

cohort 2

next week we return

for some overview and debrief look at
the project ideas

and also look forward to Future textbook
groups so feel free to fill out the um
feedback form and encourage others to do
so as well and also take a look through
some thoughts and add more thoughts on
future groups which we'll go over next
week so thank you talk to you soon